

Little Cloud: A Digital Companion App to Support Those Experiencing Pregnancy Loss

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Abstract

Pregnancy loss, encompassing miscarriage, stillbirth, and elective termination, is a deeply emotional experience with significant psychological impacts. Despite its prevalence, societal norms and cultural sensitivities often silence discussions, leaving affected individuals isolated and underserved. This study addresses the critical gap in pregnancy loss support, particularly in Taiwan, through the design and development of Xiao Yun Duo, which means “Little Cloud” in Mandarin, a mobile application that aims at providing culturally sensitive, accessible, and empathetic support. Grounded in interviews with affected women and healthcare professionals, the application integrates tailored emotional support, professional resources, and gamified therapeutic features. Key functionalities include a personalized healing journey with AR-based planting, a care calendar for post-miscarriage recovery, anonymous community chat rooms, and a professional health blog accessible to users and the public. Iterative design processes informed by user testing emphasized emotional comfort, usability, and cultural alignment. Aligned with the SDGs3 Good Health and Well-Being, Xiao Yun Duo aspires to enhance emotional well-being and societal awareness, creating a supportive ecosystem for women experiencing pregnancy loss.

CCS Concepts

• Human-centered computing; • Interaction design; Interaction design process and methods; • User interface design;

Keywords

Pregnancy loss support, Emotional design, Human-centered design, Accessibility technologies, Digital health, Cultural sensitivity, User-centered evaluation

ACM Reference Format:

Kuan-Ming Wang, You-Tong Lin, and Tsai-Ling Huang. 2025. Little Cloud: A Digital Companion App to Support Those Experiencing Pregnancy Loss. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720301>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720301>

1 INTRODUCTION

Pregnancy Loss refers to the experience of losing a fetus during pregnancy due to miscarriage, stillbirth, or elective termination, often accompanied by profound psychological and emotional impacts. [5] [12] The proportion of women experiencing pregnancy loss is significantly higher than commonly perceived by the public. [2] [3] [4] This highlights the urgent need to address and support this group. However, societal expectations of successful pregnancies and childbirth often overshadow the significance of pregnancy loss as a critical issue. Within the traditional cultural framework in Taiwan, which places strong expectations on women to successfully conceive and give birth while discouraging the open sharing of private matters like household issues [13], there is a lack of sufficient understanding of pregnancy loss, which can lead to both short-term and long-term negative impacts on individuals. [7] In Asian cultures, societal norms frequently enforce silence on the topic of miscarriage, leaving individuals who experience pregnancy loss isolated and with limited access to support resources. [11] The field of HCI has emphasized designing services tailored to the needs of specific user groups. With the increasing focus on mental health, [6] various services are now available to support pregnant women or address women's mental health more broadly. [6] Despite some exploration such as Not Alone designed by Andalibi et al. [9], there is still a lack of dedicated services specifically designed to support women who have experienced pregnancy and perinatal loss. To address this challenge, we have designed a mobile application “Xiao Yun Duo”, i.e. “Little Cloud” in Mandarin. This application focuses on three primary objectives:

1. Private Space: Provide easily accessible support to reduce the limitations and pressures of face-to-face interactions.
2. Resource Integration: Consolidate existing miscarriage-related support channels to help users gain a clearer understanding of their circumstances and access comprehensive resources.
3. Emotional Healing: Deliver gentle and practical psychological support through companionship, allowing users to receive meaningful support without requiring explicit “awareness of illness”.

2 RELATED WORK

Pregnancy loss is a complex and multifaceted issue, intertwined with cultural, medical, and psychological dimensions. [8] As societal attention to maternal health increases, technology that aims at supporting individuals through the process of pregnancy loss

has begun to emerge. We analyzed existing pregnancy loss apps and summarized their key features, focusing on emotional support, resource integration, and user-centered design principles to inform the development of our approach.

2.1 Context in Taiwan

Local governments have introduced maternal mental health welfare programs, offering subsidies to women who have experienced miscarriage for professional mental health evaluations at clinics. Private organizations, such as the association, manage anonymous online chat groups to provide long-term emotional support for women. Additionally, Taiwan has a cultural tradition of postpartum recovery, commonly referred to as *zuo yue zi* (坐月子). Postpartum care centers catering to this tradition often extend their services to include tailored recovery programs, known as “mini confinement” (*xiao yue zi*), for women recovering from miscarriage. However, there remains a significant gap in providing long-term, life-integrated psychological support for women experiencing pregnancy loss in Taiwan.

2.2 Digital Supports for Pregnancy Loss

Existing digital services that support women experiencing pregnancy loss have primarily focused on community-driven emotional support and informational resources. Three notable applications illustrate the diversity in approaches but also reveal limitations in addressing the multifaceted needs of users: *Loss to Love*¹, developed by American women who have experienced pregnancy loss, emphasizes community and group engagement by hosting various healing activities such as yoga and astrology-based therapies. While these activities provide opportunities for shared healing, the app primarily serves as a gateway to community participation and lacks features that cater to private, self-guided healing. *Pregnancy After Loss Support*², designed by a U.S.-based nonprofit organization Pregnancy After Loss Support (PALS), targets individuals who are pregnant after experiencing a miscarriage. The app offers personalized self-care tools and confidence-building messages. However, it is specifically tailored for those in subsequent pregnancies, leaving non-pregnant or non-conceiving parents without adequate support. *Loss*³, developed in Rotterdam by medical and nursing professionals, takes an educational approach. It compiles articles on physiological knowledge, poetry, and miscarriage-related content, paired with therapeutic exercises to support emotional healing. While this approach offers valuable resources, its reliance on text-based content may be less engaging or accessible for individuals who prefer more interactive or visually appealing support. In designing pregnancy loss support apps, it is crucial not only to provide knowledge and facilitate community engagement but also to address the sensory and emotional needs of users during this deeply vulnerable period. Drawing inspiration from emotional healing apps, we focused on features that align with the needs of

individuals experiencing pregnancy loss. *Emmo*’s⁴ “Daily Emotion Tracking” allows users to develop self-awareness by recording and reflecting on their emotions, providing a structured yet gentle tool for emotional exploration. Similarly, *Hearty Journal*⁵ leverages an illustrated, visually comforting style to create a warm and inviting atmosphere that fosters a sense of safety and support for those experiencing pregnancy loss—individuals particularly susceptible to emotional distress—these thoughtful approaches to emotional healing offer highly valuable references. In the context of Taiwan, pregnancy loss remains both a medical and culturally sensitive issue. Existing research has addressed the cultural aspects of pregnancy loss, but there have been few designs addressing challenges in Asian cultural context. To fill this gap, this study aims to design an application that provides culturally sensitive, tailored emotional support. The proposed app integrates professional resources, gamified therapeutic features, and private self-guided tools to create a comprehensive and accessible support system for users.

3 USER RESEARCH

To gain deeper insights, we conducted user research through public discussions, in-depth interviews with individuals who have experienced miscarriage, and consultations with professionals.

1. **Public Perception (Figure 3): General Understanding of Pregnancy Loss.** We organized campus events to assess public understanding of pregnancy loss. These events featured a discussion led by obstetricians and a hands-on healing plant workshop. A total of 33 participants, including students, faculty, and staff members from the campus community attended, including 5 individuals with pregnancy-related experiences. These sessions provided a baseline understanding of societal perceptions and attitudes toward pregnancy loss, as well as feedback on potential support mechanisms.
2. **Experiential Insights: Women with Miscarriage Experiences.** We conducted 1.5-hour in-depth interviews with 13 women who had experienced miscarriage. These conversations aimed to capture their authentic thoughts, emotions, and needs during their journey through pregnancy loss. Additionally, we observed three anonymous online chat groups on messaging platforms to understand the role and dynamics of community-based support in providing emotional relief and connection.
3. **Professional Perspectives: Healthcare and Support Providers.** To identify gaps in professional resources and support systems, we interviewed: Two nurses, two psychologists, two nutritionists, one physician, and two miscarriage support organizations.

Furthermore, we participated in an 8-hour miscarriage counseling training course organized by the Taiwan Medical and Health Counseling Psychology Association. This course provided valuable insights into the current state of professional resources and highlighted areas where support for women experiencing pregnancy loss could be improved. This comprehensive research approach allowed us to gather diverse perspectives, identify critical gaps in

¹Synappz Digital Healthcare. (2025, March 1) *Loss Handcrafted with love* <https://loss-app.info/en/>

²Pregnancy After Loss Support 2014 <https://pregnancyafterlosssupport.org/pregnancy-after-loss-app/>

³Loss to Love (2025, March 1) *Loss to Love* <https://losstolove.app/>

⁴Emmo (2025, March 1) *Emmo* Google Play [https://play.google.com/store/apps/details?id=\\$emmo.diary.app&hl=\\$zh_TW](https://play.google.com/store/apps/details?id=$emmo.diary.app&hl=$zh_TW)

⁵Hearty Journal (2025, March 1) *Hearty Journal* <https://hearty.me/>

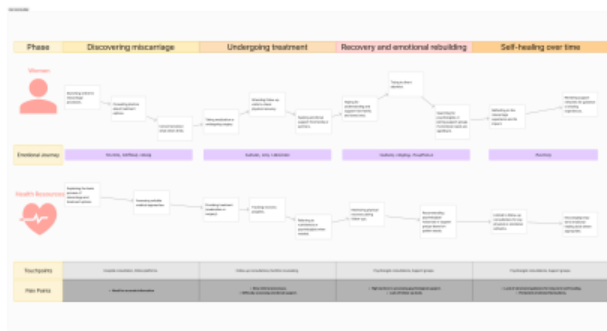


Figure 1: Experience journey map for women and professionals.

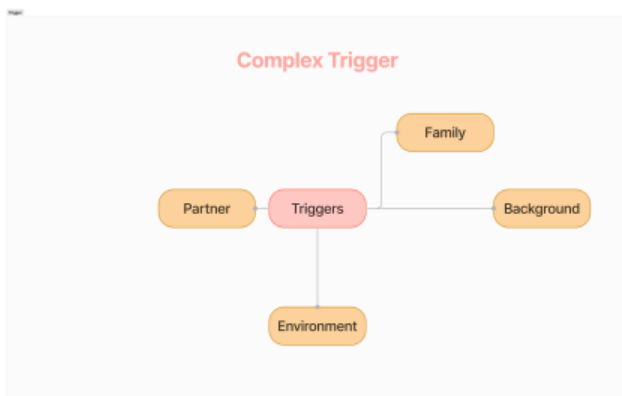


Figure 2: The mind map of triggers for pregnancy loss among women based on interviews.



Figure 3: Conducting user research through a physical discussion event.

existing support, and refine our design solutions to address the needs of this vulnerable population.

4 FINDINGS AND DESIGN IMPLICATIONS

We mapped out the experience journey and triggers of pregnancy loss based on the data we collected (Figure 1&2). These findings informed us of the development of design implications for creating a meaningful and effective support solution.

4.1 Key Insights

Women navigating pregnancy loss often encounter significant emotional and practical obstacles, compounded by a lack of understanding about miscarriage and fragmented access to resources. Community support, such as online groups, provides emotional relief but lacks professional guidance and is frequently disrupted by issues like keyword censorship and spam bots. Miscarriage hotlines, while available, are underutilized, suggesting low public awareness or accessibility barriers. Professionals involved in medical care also face limitations. Obstetricians and nurses frequently encounter emotionally distressed patients but lack specialized psychological training. This leads to inadequate support and, at times, inappropriate suggestions, such as visiting a temple for solace. Additionally, while physical recovery follows a defined timeline, the emotional toll of pregnancy loss can persist for years or resurface unexpectedly, often triggered by societal pressures or personal reflections. For example, one interviewee shared that she hid her miscarriage from her family to avoid intrusive questions but often cried alone at night. Despite this, she later expressed a desire to share her experiences to help others.

4.2 Implications for Design

To address these challenges, there is a clear need for a digital solution that offers a holistic approach, combining emotional support, professional guidance, and privacy. Such a platform should integrate fragmented resources into a cohesive system and provide a safe space for users to navigate their emotions. Moreover, incorporating interactive and gamified elements could offer long-term support that feels approachable and engaging. A digital companion app, tailored for women experiencing pregnancy loss, emerges as an ideal solution. Its features can address the dual need for privacy and accessibility, allowing users to engage with the platform at their own pace. In Taiwan, where digital service penetration is exceptionally high, this approach ensures widespread reach and usability. By offering tools like mood journaling, resource integration, and interactive healing activities, the app can empower users to take control of their emotional recovery in a way that feels both safe and supportive.

The findings underscore the need to bridge the existing gaps in support systems.

Our aim is to provide accessible, personalized support during the critical early stages of recovery. With high global internet and smartphone usage, especially in Taiwan, where over 80% of the population own a smartphone, even young, underage girls can easily access the app.

1. With the phone always by their side, the app provides constant, daily reminders and support, offering users a way to stay connected with their mental and physical well-being, anytime and anywhere.
2. When interacting with others in person, sometimes it's difficult to find the right words to comfort someone. The app removes the pressure and limitations of face-to-face interactions, allowing users to focus on self-healing in a private, safe space without external noise.



Figure 4: Application user interface changes.

5 DESIGN PROCESS

We adopted an iterative design process in collaboration with partner organizations. After each round of design and user testing, we discussed the results with organization members to inform the next phase of design iterations.

5.1 Initial Ideation

The design process began by conceptualizing key features rooted in companionship, healing, physical support, and community to address the unique needs of women experiencing pregnancy loss. These features were carefully crafted to create an empathetic and supportive digital experience. The initial design included the following core elements (Figure 4): Companion Character 'Little Soul': A virtual companion that offers daily comforting messages and emotional support to users throughout their recovery journey. Emotional Tracking with 'Moon Diary': A feature that enables users to track their emotions using moon-phase icons, providing a gentle and intuitive way to reflect on their feelings over time. Plant Nurturing Inspired by Game Therapy: A gamified planting experience where users can nurture virtual plants, symbolizing growth and healing, fostering emotional resilience. Personalized Recovery Calendar: A tailored calendar that offers step-by-step guidance for postpartum recovery, blending traditional health practices with modern insights. Chatrooms for Mutual Support: Anonymous chat spaces where users can connect with others who share similar experiences, fostering a sense of community and shared understanding. The design prioritized clarity, emotional comfort, and user-friendliness. Complex or fragmented interface elements were removed after consultation with professional associations to ensure the app remained intuitive and approachable, aligning with its therapeutic objectives.

5.2 First Prototype

We conducted the first round of testing for Xiao Yun Duo's prototype with 11 individuals who had experienced pregnancy loss. Each session lasted one and a half hours and provided participants with the opportunity to explore the app's features through an interactive Figma prototype (Figure 5). During these sessions, participants navigated key features, including the virtual plant interaction, a postpartum recovery calendar, and expert knowledge resources. They were encouraged to share their feedback and discuss how the app could better support their emotional and practical needs.

Key Feedback and Improvements:

1. **Renaming 'Little Soul' to 'Little Cloud':** Participants shared that the term "Little Soul" evoked sad memories or discomfort, as it recalled the loss they had experienced. To make the app feel more comforting and approachable, we renamed the feature to "Little Cloud," which participants described as gentle, cute, and emotionally soothing.
2. **Enhancing the Plant Care Feature with AR-Based Plants.** The original plant care feature was replaced with AR-based plants to create a more interactive and lifelike experience. This change made the feature feel closer to participants' daily lives, providing a more engaging and relatable way to visualize emotional healing and growth.
3. **Introducing Expert-Written Articles.** To address the gap in knowledge support, we added a dedicated section for expert-written articles. These resources were designed to provide clear, practical information on mental and physical health topics relevant to pregnancy loss. Additionally, this feature serves as a bridge, helping users connect with professional support, such as psychologists and healthcare providers, for deeper assistance when needed.

These refinements not only made the app more emotionally sensitive but also strengthened its practical value by offering personalized and professional support. The addition of AR-based features and expert resources ensures that users feel guided and supported throughout their healing journey.

5.3 Second Prototype

Following the initial round of feedback and adjustments, we collaborated with professionals in physical and mental health for a more extensive test of the app. Second Prototype LINE (Figure 6) LINE Test and LINE concept validation. This phase involved running participants through a full 30-day process, allowing us to gather more in-depth and long-term feedback on user engagement and emotional resonance. During this round of testing, participants interacted with the newly refined features, which were specifically designed to enhance emotional comfort, foster connection, and create a more personalized experience.

Key Feedback and Improvements:

Soft, Calming Design. Feedback from participants emphasized the importance of a visually calming and emotionally supportive design.

1. softer color scheme and relaxation features. The homepage's cool blue tones were replaced with warm pinks to evoke safety and warmth, while the "Little Cloud" edges were softened to pastel blue for a more soothing aesthetic. Natural white noise, including sounds like waves and gentle rain, was added to aid relaxation, sleep, and a sense of privacy.
2. Personalized interface. A personal profile page was introduced, allowing users to customize their avatar and feel more connected to the application. The interface was tailored with user-specific data and personalized recommendations, reinforcing the app's empathetic design.
3. Expanded AR features: The AR plant feature evolved into a "Planet Garden," enabling users to cultivate a digital healing space tied to milestones and emotional progress. Interactive



Figure 5: First prototype with 6-3.

storytelling unlocked narratives as plants grew, symbolizing personal transformation and growth.

4. Tailored chatroom categories. The chatroom was divided into “Miscarriage Healing” for emotional support and “Trying to Conceive Nest” for those preparing to try again. This segmentation helped address specific needs without overwhelming users.

By the end of the 30-day testing process, participants reported feeling more connected to the app and expressed increased trust in its ability to guide them through difficult emotional experiences. This feedback will serve as the foundation for the next iteration of Xiao Yun Duo, ensuring it continues to meet the needs of its users.

6 FINAL DESIGN

6.1 Service Overview

The final version of the application integrates feedback from previous tests and usability heuristic principles. Developed using Unity, the final prototype includes four main features to achieve the design goals:

Feature 1: Personalized Healing(7) Upon logging into the application, users are asked for their consent to record basic information related to their miscarriage. On the app’s homepage, a character called “Little Cloud” provides daily comforting messages and reminders to water the virtual plant. The plant-watering process requires users to open their phone camera to water the plant and observe its growth through different stages. Users can record their moods in a Mood Journal using text and moon phase icons, allowing them to reflect on their emotions over time. The changing moon phase icons also influence the color and appearance of the plant’s growth. After completing a 30-day journey, each user will

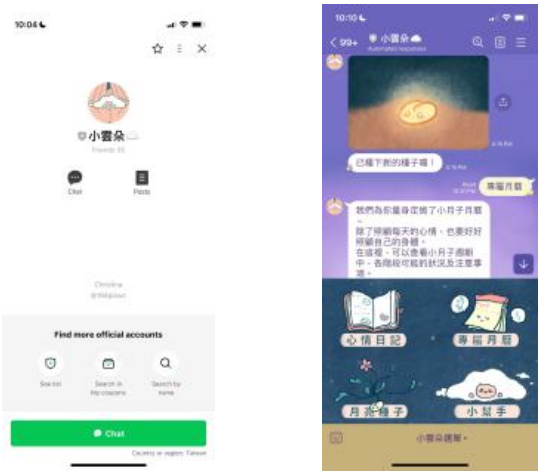


Figure 6: Second prototype with Line Account.

cultivate a unique plant that reflects their experience. Users can continue planting beyond the initial 30-day recovery period, creating an ongoing opportunity for emotional and personal growth.

Feature 2: Health Professional Platform. The Care Calendar simplified traditional Chinese medicine (TCM) recovery guidance, offering tailored post-miscarriage recovery recommendations based on user-provided information. Through the Health Blog, users can access short articles shared by professionals. They can use the keyword search feature to find resources that address their personal concerns. For advanced medical consultation needs, users can utilize the Contact Us function to connect with the appropriate

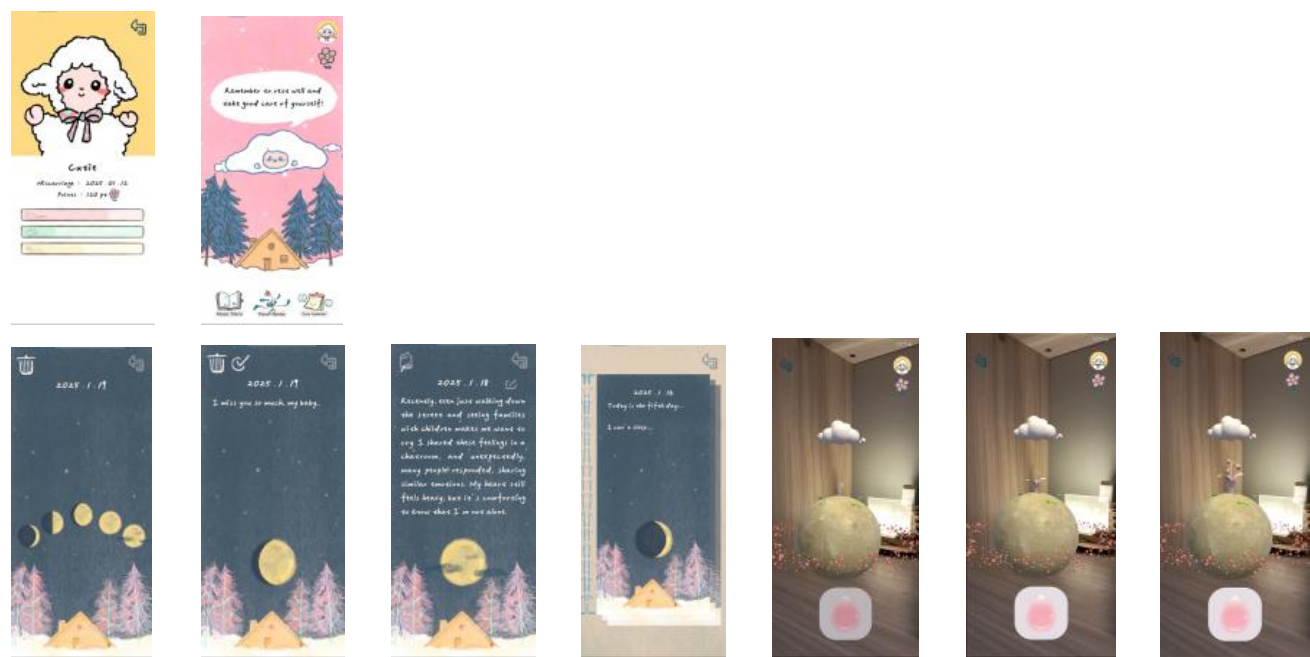


Figure 7: 7-1 to 7-10

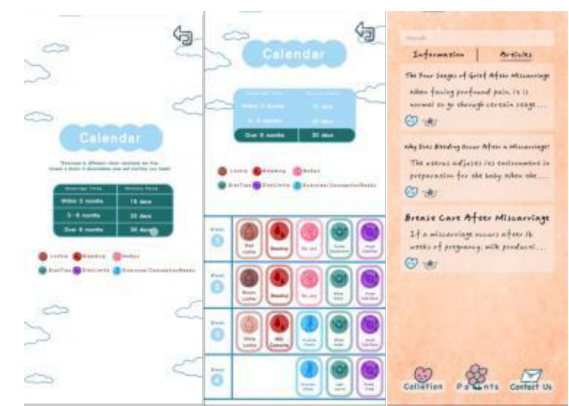


Figure 8: 8-1 to 8-3

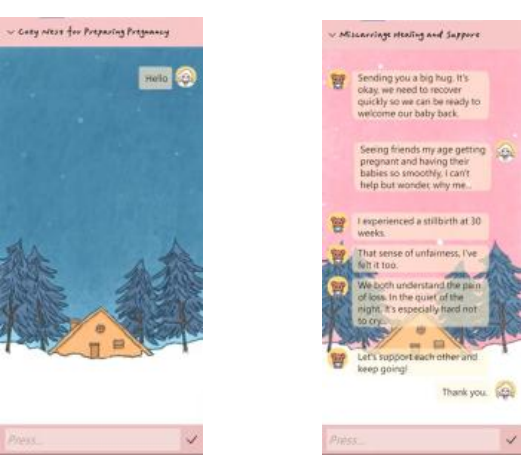


Figure 9: 9-1 to 9-2

professional organizations. Articles updated on the app are also accessible to the public via a web version, helping to raise awareness and understanding of pregnancy loss. This feature aims to reduce the barriers for users to actively seek help, addressing both individual and societal challenges.

Feature 3: Anonymous Support Community. The Anonymous Community Chatrooms provide a safe space for users to connect and support one another. These chat rooms are divided into Miscarriage Communities and Conception Planning Communities, ensuring users can interact in the most comfortable environment. Additionally, the platform minimizes disruptions from chatbots and avoids restrictions on discussing sensitive topics.

6.2 User journey

Through its personalized features like anonymous chatrooms, expert articles, care calendars, and emotional tracking tools, the Little Cloud App provides a safe platform for emotional expression for women experiencing pregnancy loss, effectively aiding in the management and recovery of their mental and physical health. The use of the app facilitates understanding and acceptance of their personal experiences, gradually helping them to overcome their grief. Moreover, by recommending the app to other women with similar experiences, users become companions in healing, thereby

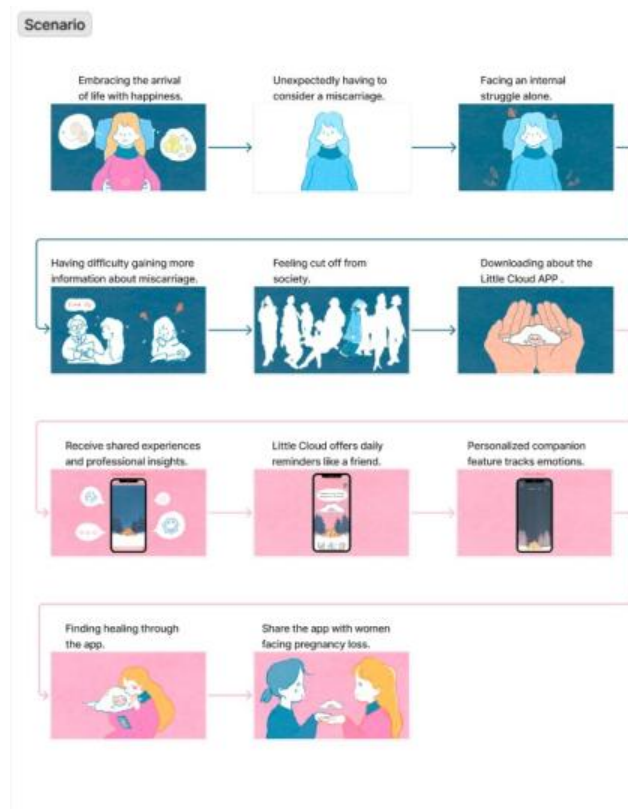


Figure 10: The user journey of the Little Cloud App with women experiencing pregnancy loss.

extending Little Cloud App's practical effectiveness in providing psychological support and fostering community resonance.

7 LIMITATION AND FUTURE RESEARCH DIRECTION

This study faced several limitations, primarily stemming from the testing process and sample composition. While the prototype was evaluated by individuals with pregnancy and postpartum experiences, we did not include women actively undergoing pregnancy loss at this stage of development. This decision was made to protect the well-being of this highly vulnerable population. Instead, we focused on conducting in-depth interviews with women who had previously experienced pregnancy loss. These participants provided valuable insights into their emotional and psychological needs, helping to guide the design in a sensitive and ethical manner. Engaging directly with individuals actively experiencing pregnancy loss will be a priority in future iterations once Little Cloud has undergone further refinement to ensure its features are both supportive and non-intrusive. Looking ahead, future research and development will prioritize targeted testing with women actively experiencing pregnancy loss, allowing the design to better align with their real-time needs. Collaborations with professional psychological and medical organizations will be strengthened to ensure the integration of validated therapeutic practices and resources.

Usability refinement will continue through iterative feedback cycles to ensure the app remains intuitive and effective, particularly for individuals navigating the sensitive and complex emotions associated with pregnancy loss. By addressing these areas, the Little Cloud app can evolve into a more impactful tool, providing culturally sensitive and deeply empathetic support for women during one of the most challenging periods of their lives.

Acknowledgments

We would like to express our sincere gratitude to our advisor, Joey Chiao-Yin Hsiao, for invaluable guidance and support. We are also deeply thankful to the Taiwan Medical and Health Counseling Psychology Association for their assistance and insightful recommendations, which greatly contributed to the development and refinement of our work.

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